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Advancing Cybersecurity Practice: Explainable Machine Learning for Network Intrusion Detection

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Abstract

This research investigates explainable artificial intelligence (XAI) integration within machine learning (ML)-based intrusion detection systems (IDS), focusing on distinguishing malicious from benign network activities. We employed Random Forest and XGBoost models evaluated on widely recognized datasets, including NSL-KDD and UNSW-NB15, using both binary and multi-class classification tasks. The objective was to enhance cybersecurity operations through improved model transparency and interpretability. By integrating SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations), the study offers comprehensive global and local insights into model decision-making processes. Results demonstrate SHAP's effectiveness in providing a broad, dataset-wide understanding of feature interactions and importance, while LIME facilitates targeted analysis of specific misclassified instances, thereby identifying potential biases. This dual approach advances the state-of-the-art by enabling security analysts to effectively interpret AI-driven decisions, reduce false positives, and refine response strategies. The integration of XAI into IDS frameworks significantly improves system reliability, transparency, and practical applicability in real-world cybersecurity operations. Our findings provide actionable insights for security practitioners, demonstrating how explainable AI can enhance operational decision-making and improve the effectiveness of intrusion detection systems in production environments. The study also presents a practical platform that enables security teams to interact with and interpret model predictions, bridging the gap between theoretical research and operational cybersecurity practice.

Keywords

Explainable AI (XAI), Intrusion Detection System (IDS), Machine Learning in Cybersecurity, Network Threat Detection

Advancing Cybersecurity Practice: Explainable Machine Learning for Network Intrusion Detection

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***Abstract*—This research investigates explainable artificial intelligence (XAI) integration within machine learning (ML)-based intrusion detection systems (IDS), focusing on distinguishing malicious from benign network activities. We employed Random Forest and XGBoost models evaluated on widely recognized datasets, including NSL-KDD and UNSW-NB15, using both binary and multi-class classification tasks. The objective was to enhance cybersecurity operations through improved model transparency and interpretability. By integrating SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations), the study offers comprehensive global and local insights into model decision-making processes. Results demonstrate SHAP's effectiveness in providing a broad, dataset-wide understanding of feature interactions and importance, while LIME facilitates targeted analysis of specific misclassified instances, thereby identifying potential biases. This dual approach advances the state-of-the-art by enabling security analysts to effectively interpret AI-driven decisions, reduce false positives, and refine response strategies. The integration of XAI into IDS frameworks significantly improves system reliability, transparency, and practical applicability in real-world cybersecurity operations. Our findings provide actionable insights for security practitioners, demonstrating how explainable AI can enhance operational decision-making and improve the effectiveness of intrusion detection systems in production environments. The study also presents a practical platform that enables security teams to interact with and interpret model predictions, bridging the gap between theoretical research and operational cybersecurity practice.**

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I. Introduction

Distinguishing between normal and abnormal network traffic behavior is critical in cybersecurity operations. AI/ML models like Random Forest, XGBoost, and ensemble meta-models effectively analyze network traffic to detect cyber threats, leveraging structured security data to enhance feature identification and decision-making. AI/ML-driven intrusion detection has been widely studied [1], demonstrating its superiority over traditional statistical anomaly detection and rule-based systems through several key advantages: (1) automated adaptation to evolving threats without manual rule updates, (2) real-time processing capabilities that can handle large volumes of network traffic, (3) improved detection accuracy through pattern recognition in complex data, and (4) scalability to process heterogeneous network data efficiently. AI enables automated, adaptive detection with improved accuracy, addressing evolving cyber threats more efficiently than traditional approaches.

Several studies have explored decision-tree-based models, such as random forest and XGBoost, to enhance intrusion detection. Collectively, these works [2–7] demonstrate that while AI/ML models show promise in cybersecurity applications, significant gaps remain in explainability and interpretability. Previous research has shown that ensemble learning approaches can improve detection accuracy [2], and AI-driven models can effectively target zero-day attacks [3]. However, these approaches often lack comprehensive explainability frameworks, limiting practitioners' ability to understand and trust model decisions. Recent work has highlighted the importance of robust explainability frameworks [4] and the need to address interpretability in practical deployments [5, 6]. Based on these findings, there is a clear need for frameworks that provide both global and local explanations, enabling security analysts to understand both dataset-wide patterns and instance-specific decisions.

Our study builds upon these works by integrating SHAP and LIME to provide a comprehensive understanding of feature influence at both local and global levels. We selected SHAP (specifically TreeSHAP) for its ability to provide global, dataset-wide feature importance insights optimized for tree-based models like Random Forest and XGBoost, enabling security analysts to understand overall model behavior and feature interactions. LIME was chosen for its model-agnostic nature and ability to provide local, instance-specific explanations that help analysts investigate individual alerts and validate model decisions against domain knowledge. This dual approach addresses the limitations of prior work by offering both global pattern recognition and local case analysis, enabling a more comprehensive understanding of model decision-making. Unlike prior research, which evaluates XAI techniques in isolation, we present a comparative analysis to offer security analysts a clearer perspective on model decision-making.

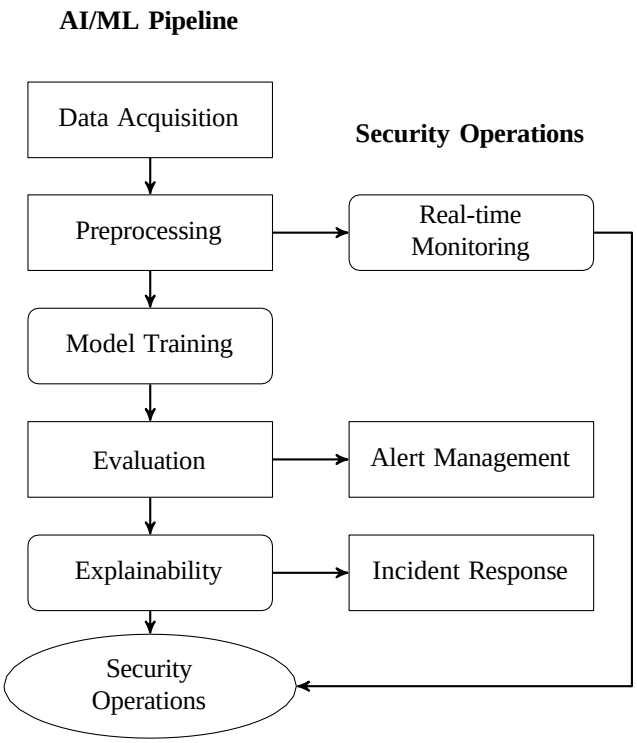

Fig. 1: Explainable Intrusion Detection workflow with practical security operations integration

This paper presents a cohesive framework that integrates benchmark datasets, machine learning-based intrusion detection, and XAI techniques to enhance the detection and interpretation of normal versus abnormal activity in cybersecurity. Our approach leverages SHAP and LIME to provide both global and local explainability, offering security analysts deeper insights into model decisions. Fig. 1 provides an overview of this workflow, illustrating the sequential processes from data preprocessing to explainability.

The workflow, as illustrated in Fig. 1, demonstrates how our framework integrates with real-world security operations. The left side of the figure shows the core AI/ML pipeline, while the right side highlights practical security operations applications. This integration enables security teams to:

- Monitor network traffic in real-time using preprocessed data
- Manage alerts based on model evaluation results
- Respond to incidents with explainable model insights

This practical integration ensures that our research directly contributes to improving security operations by providing transparent and interpretable AI-driven insights for cybersecurity professionals.

II. Literature Review

A. ML for Intrusion Detection

ML has revolutionized cybersecurity operations, particularly in IDS, by offering robust and scalable solutions to identify malicious activities [1, 8–10]. Common ML techniques include Random Forest, XGBoost, and deep learning models, all demonstrating superior performance over traditional statistical anomaly detection and rule-based methods. RF and XGBoost, known for handling large, heterogeneous datasets efficiently, offer high accuracy and low computational overhead compared to neural networks [3]. Here, "efficiency" refers to three key aspects: (1) computational efficiency—faster training and inference times compared to deep learning models, enabling real-time processing in operational environments; (2) memory efficiency—lower memory requirements making them suitable for resource-constrained deployments; and (3) preprocessing efficiency—the ability to handle mixed data types (categorical and numerical) with minimal preprocessing compared to neural networks that require extensive feature engineering. However, ML-based systems require extensive preprocessing and hyperparameter tuning to avoid overfitting and maintain generalization capabilities [11]. In operational environments, these models must balance detection accuracy with computational efficiency to ensure real-time threat detection and response.

B. XAI in Cybersecurity

XAI is critical for enhancing trust and interpretability of AI systems in cybersecurity operations [12–15]. Techniques like SHAP and LIME are widely employed to provide transparency in AI-driven decisions. SHAP provides global explanations, enabling security analysts to understand feature influences across datasets, whereas LIME offers local explanations, providing insights into individual predictions [6]. Despite their effectiveness, challenges remain in applying XAI to cybersecurity operations, particularly concerning computational complexity and potential vulnerability to adversarial manipulations [4]. For example, adversarial attacks on XAI methods can manipulate explanations without changing predictions, potentially misleading security analysts into trusting incorrect model decisions [4, 15]. Such vulnerabilities highlight the need for robust explainability frameworks that can detect and mitigate adversarial manipulation. Recent work has focused on making XAI techniques more practical for security operations centers (SOCs), where analysts need quick, actionable insights to respond to threats effectively.

C. Benchmark Datasets for Network Intrusion Detection

Benchmark datasets such as NSL-KDD [16] and UNSW-NB15 [17] play pivotal roles in evaluating intrusion detection methods. NSL-KDD addresses some inherent problems of its predecessor, KDD Cup 1999 [18], such as redundant data and imbalance issues, yet still faces criticism for not fully representing modern threats. UNSW-NB15 includes contemporary attack scenarios and offers diverse network behaviors, making it suitable for current cybersecurity research. Nevertheless, both datasets pose challenges related to class imbalance, data distribution shifts, and the evolving nature of cybersecurity threats, affecting model generalizability [5]. These challenges are particularly relevant in operational environments where

models must adapt to new attack patterns and maintain high detection rates with minimal false positives.

D. Adversarial Robustness and Resilience in AI-driven Cybersecurity Systems

The adversarial robustness of ML models in cybersecurity operations is an increasingly critical research area due to rising threats from adversarial attacks aimed at manipulating model predictions [19–22]. Adversarial examples exploit vulnerabilities in the model's decision-making processes, causing significant disruptions in IDS effectiveness. Recent literature focuses on defense mechanisms such as adversarial training, input transformation, and robust optimization techniques, aiming to enhance resilience against these threats [4, 7]. Despite these advances, ensuring adversarial resilience remains challenging due to constantly evolving attack vectors and the dynamic cybersecurity landscape. In operational settings, these challenges are compounded by the need for real-time detection and response, requiring models to be both robust and efficient.

E. Practical Implementation Challenges

Recent research has highlighted several practical challenges in deploying ML-based IDS in operational environments [1, 10, 13]. These include:

- Real-time processing requirements and computational resource constraints
- Integration with existing security infrastructure and workflows
- Model maintenance and updates in response to evolving threats
- Balancing detection accuracy with operational efficiency
- Ensuring model interpretability for security analysts

These challenges underscore the need for practical, operationally-focused research that addresses the real-world constraints and requirements of cybersecurity operations. Our work contributes to this area by providing a framework that balances technical innovation with practical applicability. Specifically, our framework balances technical innovation (through advanced XAI integration using both SHAP and LIME, comparative analysis of global and local explanations, and sophisticated feature interaction analysis) with practical applicability (through real-time processing capabilities, containerized deployment using Docker, REST API interfaces for easy integration, and mechanisms for automated model updates and monitoring). This dual focus ensures that theoretical advances in explainable AI translate directly to operational use, enabling security teams to leverage cutting-edge interpretability techniques while maintaining the efficiency and reliability required in production environments.

III. Methodology

The primary models selected for this research were Random Forest and XGBoost, chosen for their practical advantages in operational environments [3, 11]. Random Forest, an ensemble of decision trees, was selected for its resilience and reduced propensity to overfit through averaging predictions across multiple decorrelated trees [11]. Its ability to naturally handle categorical and continuous features with minimal preprocessing makes it suitable for high-dimensional network traffic data, where a variety of protocol, service, and statistical features must be integrated. XGBoost (eXtreme Gradient Boosting) was incorporated for its high-performing approach in handling tabular data and its ability to capture complex, non-linear patterns by iteratively fitting new trees to address the residual errors of prior iterations [3]. Additionally, both XGBoost and Random Forest are computationally efficient compared to deep learning models, making them more practical for large-scale intrusion detection where real-time analysis and resource constraints are important considerations.

A critical component of the methodology involved data preprocessing to transform all categorical attributes, such as protocol type, service, and flags, into numerical representations. Label encoding was used to map categorical values to integer codes, ensuring that the algorithms could process them effectively. The datasets used in this study, UNSW-NB15 and NSL-KDD, come pre-split into training and testing sets, eliminating the need for additional partitioning. We utilize these predefined splits to evaluate model performance without modifying the original dataset structure. The UNSW-NB15 dataset was divided into approximately 68% training (175,341 instances) and 32% testing (82,332 instances), while approximately 85% of the NSL-KDD dataset (125,973 instances) was allocated for training and 15% (22,544 instances) for testing. Since these datasets were pre-cleaned before being split, no additional data imputation was required. Feature encoding was applied during preprocessing to convert categorical variables into numerical representations, ensuring compatibility with machine learning models.

A. Operational Considerations

To ensure practical applicability in real-world security operations, we implemented several key considerations:

- Real-time Processing: Models were optimized for efficient inference, ensuring they can process network traffic in real-time without significant latency.
- Resource Efficiency: Both Random Forest and XGBoost were configured to balance model complexity with computational requirements, making them suitable for deployment in resource-constrained environments.
- Model Maintenance: The framework includes mechanisms for periodic model updates and retraining to adapt to evolving threats.
- Integration Capabilities: The system was designed to easily integrate with existing security infrastructure and workflows.

B. Model Selection and Hyperparameter Tuning

Hyperparameter tuning was conducted using Bayesian Optimization, which is particularly suitable for operational environments due to its efficiency in finding optimal configurations [11]. Rather than exhaustively searching the hyper-

parameter space, as is the case with grid search, or randomly sampling it, Bayesian Optimization employs a surrogate model to locate promising regions more efficiently. This approach has been shown to be more effective than traditional methods for hyperparameter optimization in machine learning applications [11]. This not only reduces computational overhead but also accelerates convergence toward near-optimal parameter configurations for both Random Forest and XGBoost. Typical hyperparameters optimized included:

- XGBoost: Number of estimators, Maximum depth, Learning rate, Subsample ratio, Column subsample ratio
- Random Forest: Number of trees, Maximum features, Minimum samples required for split, Maximum depth, Minimum samples per leaf

These parameters were optimized with a focus on both model performance and operational efficiency, ensuring the models can be effectively deployed in production environments.

C. Implementation Framework

The implementation framework was designed with operational deployment in mind:

1) Data Pipeline:
   - Real-time data ingestion and preprocessing
   - Feature extraction and transformation
   - Batch processing for model updates
2) Model Deployment:
   - Containerized deployment using Docker
   - REST API interface for model inference
   - Integration with security monitoring systems
3) Monitoring and Maintenance:
   - Performance metrics tracking
   - Automated model retraining triggers
   - Alert management and incident response integration

Each framework component specifically supports model efficiency: (1) The data pipeline enables low-latency processing through real-time ingestion and optimized preprocessing, reducing the time from data arrival to prediction; (2) Containerized deployment ensures consistent performance across environments and enables horizontal scaling to handle increased traffic loads; (3) The REST API interface provides efficient, standardized access to model predictions without requiring complex integration; and (4) Automated monitoring allows for proactive optimization by identifying performance degradation before it impacts operations. This framework ensures that the models can be effectively deployed and maintained in operational security environments while providing the necessary explainability features for security analysts.

D. Dataset Selection and Preprocessing

We began by selecting benchmark datasets that include both benign and malicious network traffic. For this study, we utilized the NSL-KDD and UNSW-NB15 datasets. These datasets are widely recognized in cybersecurity research [16–18] and come preprocessed with training and testing splits, making them ideal for evaluating AI/ML models. As shown in Tables I and II, their variety of attack types provided a solid foundation for testing binary and multi-class classification approaches.

Before feeding the data into the models, we applied the following preprocessing steps:

1) Categorical Encoding: Categorical features were transformed into numerical values to ensure compatibility with the machine learning models.
2) Class Distribution Considerations: While some attack types were underrepresented in the dataset, the models demonstrated strong classification performance without requiring explicit class weighting techniques. However, some misclassifications still occurred, particularly in distinguishing certain attack types. These instances are further examined using XAI techniques to better understand model decision-making and potential areas for improvement.

E. Binary and Multi-Class Classification

We explored both binary and multi-class classification to evaluate their effectiveness in distinguishing between normal and malicious behavior.

Binary classification provides a high-level distinction between normal and malicious traffic. This approach is valuable for scenarios where the primary goal is to quickly flag any abnormal behavior for further investigation. In this study, we employ binary classification in two ways: (1) distinguishing normal traffic from all attack types combined, providing a broad anomaly detection capability, and (2) isolating specific attack types, such as Normal vs. DoS, to better understand targeted threat behaviors. For binary classification, the model was trained using supervised learning on labeled data where normal traffic instances were labeled as one class and attack traffic instances (either all attacks combined or specific attack types like DoS) were labeled as the other class. The model learns to distinguish between these classes by identifying feature patterns associated with each class through the training process. Specifically, for targeted threat behaviors (e.g., Normal vs. DoS), the model was trained on labeled instances of normal traffic and DoS attack traffic, allowing it to learn the distinguishing characteristics of DoS attacks such as high connection rates, specific protocol patterns, and traffic volume anomalies. By training the model to identify deviations from normal behavior, we ensured it could generalize well to unseen attack types, improving its real-world applicability.

F. Multi-Class Classification (Specific Attack Types)

In addition to binary classification, we utilized multi-class classification to distinguish between different types of attacks. By classifying normal traffic alongside specific attack categories, such as Denial-of-Service (DoS) and Reconnaissance, our models provide a more granular understanding of network

TABLE I: NSL-KDD Dataset Overview

Dataset	Normal	DoS	Probe	R2L	U2R
NSL-KDD (Train)	67,343	45,927	11,656	995	52
NSL-KDD (Test)	9,711	7,167	2,421	2,423	198

TABLE II: UNSW-NB15 Dataset Overview

Dataset	Normal	DoS	Reconnaissance	Exploits	Worms	Backdoor	Shellcode	Fuzzers	Generic	Analysis
UNSW-NB15 (Train)	56,000	12,264	10,491	33,393	130	1,746	1,133	18,184	40,000	2,000
UNSW-NB15 (Test)	37,000	4,089	3,496	11,132	44	583	378	6,062	18,871	677

threats. This approach allows for improved threat differentiation, aiding in the identification of distinct malicious behaviors.

Multi-class classification enables the model to recognize specific attack types rather than treating all malicious traffic as a single category. This approach is particularly valuable for distinguishing between attack types that share certain behavioral characteristics but serve different purposes, such as Denial-of-Service (DoS) and Reconnaissance. By training the model to classify multiple attack types simultaneously, we allow it to learn the nuanced differences between threats, improving its generalization to unseen attacks.

Unlike multiple binary classifiers that only compare one attack type against normal traffic, a multi-class approach provides a more comprehensive view of network threats. Each attack type exhibits unique feature interactions, and analyzing these distinctions helps validate the model's effectiveness while refining its ability to classify evolving threats. This deeper level of understanding enhances both detection accuracy and response strategies.

By implementing both approaches, we demonstrated the model's ability to:

1) Detect malicious behavior broadly by distinguishing between normal and attack traffic in a binary classification setting, where all attack types are grouped together.
2) Identify specific attack types through both binary and multi-class classification. In a focused binary setup, we differentiate between Normal and DoS traffic, while in a multi-class setting, the model learns to classify Normal, DoS, and Reconnaissance attacks separately.

G. XAI Integration

To enhance the interpretability of the models, we integrated two widely used XAI methods, LIME and SHAP:

- LIME: LIME [6] provides local explanations by analyzing individual predictions. It identifies the features that influenced a specific decision, offering detailed insights into why the model classified an instance as malicious or as a particular attack type. For example, if the model flagged an instance as a denial-of-service attack, LIME helps pinpoint the features that led to that decision. This method enables analysts to inspect specific cases and validate whether the model's decision aligns with domain knowledge.
- SHAP: TreeSHAP [6, 12], a specialized SHAP variant optimized for tree-based models, was used to explain feature contributions in Random Forest and XGBoost. Unlike traditional SHAP methods, TreeSHAP efficiently computes feature importance in decision trees, utilizing their structure for faster and more accurate contributions. SHAP provides global explanations by analyzing feature impact across the entire dataset, highlighting which factors influence the model's overall behavior. This helps analysts identify broader patterns and refine detection strategies accordingly.
- Local vs. Global Explanations: Local explanations, like those from LIME [6], focus on individual predictions, allowing analysts to inspect case-specific justifications. In contrast, global explanations, like those from SHAP [6, 12], provide an overarching view of feature importance across the entire dataset. While local methods help understand specific anomalies, global methods reveal trends and systemic behaviors that can improve model tuning and policy adjustments.
- Model-Agnostic Definition: A model-agnostic explanation method is one that can be applied to any type of machine learning model without requiring modifications to its internal structure [6, 13]. LIME is model-agnostic, meaning it can provide explanations for diverse models, including decision trees, neural networks, and ensemble methods. This flexibility makes model-agnostic approaches valuable for ensuring transparency and interpretability across various AI architectures.

This combination of local and global interpretability bridges the gap between understanding specific predictions and identifying dataset-wide trends. These insights make the model more transparent and meaningful for cybersecurity applications.

H. Evaluation Metrics

To assess the performance of our intrusion detection model, we employed multiple evaluation metrics. The confusion matrix offers a detailed breakdown of true positives, true negatives, false positives, and false negatives. Precision and recall measure the model's effectiveness in identifying malicious activities, with precision indicating the proportion of correctly predicted attacks and recall capturing the percentage of actual attacks detected. Additionally, we utilized the Receiver Operating Characteristic (ROC) curve and the

Area Under the Curve (AUC) value to evaluate the model's ability to distinguish between normal and malicious traffic. The Precision-Recall curve further highlights performance in handling imbalanced datasets, ensuring robust anomaly detection.

IV. Real-World Implications

Using both binary and multi-class classification, along with LIME and SHAP, improves real-world cybersecurity by making AI-based intrusion detection more accurate and interpretable:

1) Binary Classification: Used for both broad anomaly detection (distinguishing normal from any attack) and targeted attack differentiation (e.g., Normal vs. DoS). This allows analysts to quickly flag malicious activity while also identifying specific threats.
2) Multi-Class Classification: Provides deeper insights into distinct attack categories, allowing the model to:
   - Learn the nuanced differences between attack types, such as DoS and Reconnaissance.
   - Improve detection strategies by understanding attack-specific behaviors.
   - Reduce false positives by analyzing feature interactions and refining model predictions.

To effectively implement and use this framework in operational settings, cybersecurity practitioners and security analysts require training in several key areas. First, analysts should be trained to interpret SHAP global feature importance plots to understand overall model behavior and identify which network features are most influential across the dataset. This enables them to recognize systemic patterns and adjust detection strategies accordingly. Second, practitioners need hands-on experience using LIME explanations to investigate specific alerts, understanding how to read feature attributions and validate model decisions against their domain knowledge. Third, training should cover how to integrate XAI insights into incident response workflows, including when to trust model predictions based on explanation quality and how to use explanations to reduce false positives. Our practical platform implementation facilitates this training by providing an interactive environment where practitioners can query the model, generate explanations, and explore different scenarios. This hands-on approach helps bridge the gap between theoretical understanding and practical application, enabling security teams to effectively leverage explainable AI in their daily operations.

By demonstrating how XAI tools enhance AI/ML models, this study highlights the importance of transparency and interpretability in identifying malicious behavior and supporting real-world cybersecurity efforts.

V. NSL-KDD Binary Classification and Explainability

This section presents the results of binary classification for the NSL-KDD dataset, where the objective is to distinguish between normal and malicious traffic. We evaluate two machine learning models, Random Forest (RF) and XGBoost, using Precision-Recall (PR) curves to assess performance under imbalanced conditions. The PR curves, shown in Figure 2, provide insight into the models' ability to maintain high precision while effectively capturing true positive instances of malicious activity. To enhance interpretability, we employ TreeSHAP for global feature importance analysis using the XGBoost model. As illustrated in Figure 3, SHAP values provide insights into how each feature contributes to the model's decision-making. The red and blue dots in the SHAP summary plot represent the direction and magnitude of feature impact on predictions. Red indicates higher feature values contributing positively to classifying an instance as malicious, while blue signifies lower feature values reducing the likelihood of a malicious classification. This contrast allows for a clearer interpretation of key features influencing the model's decision-making process.

From the SHAP summary plot, we observe that source bytes (src_bytes) and destination bytes (dst_bytes) play a critical role in distinguishing between normal and malicious traffic. Additionally, features such as the number of connections to the same service (dst_host_srv_count), total connection count (count), and the destination host's same service rate (dst_host_same_srv_rate) emerge as highly influential, highlighting network traffic patterns associated with attack behavior. The service type (service) also shows strong relevance, indicating that certain services may be more susceptible to attacks.

By analyzing these explanations, security analysts can identify which features drive malicious classifications and how variations in feature values affect model outputs. The combination of performance evaluation and explainability supports a more robust understanding of machine learning-based intrusion detection in real-world cybersecurity applications.

VI. Classifying Denial-of-Service (DoS) Attacks Using XAI

The classification of Denial-of-Service (DoS) attacks versus normal traffic is examined using the NSL-KDD dataset, employing XGBoost for detection and LIME for explainability. To improve class balance, all DoS attack types—Smurf, Neptune, Pod, Teardrop, and Land—were consolidated into a single category, allowing the model to focus on distinguishing malicious activity from benign network behavior.

Figure 4 presents the evaluation of the XGBoost-based intrusion detection model for DoS attack classification using the KDD dataset, including the Precision-Recall (PR) curve, the confusion matrix, and the Receiver Operating Characteristic (ROC) curve. The PR curve, shown in the left panel, reports an Average Precision (AP) score of 0.97, demonstrating strong classification performance. The right panel displays the confusion matrix, which provides insight into the model's classification results.

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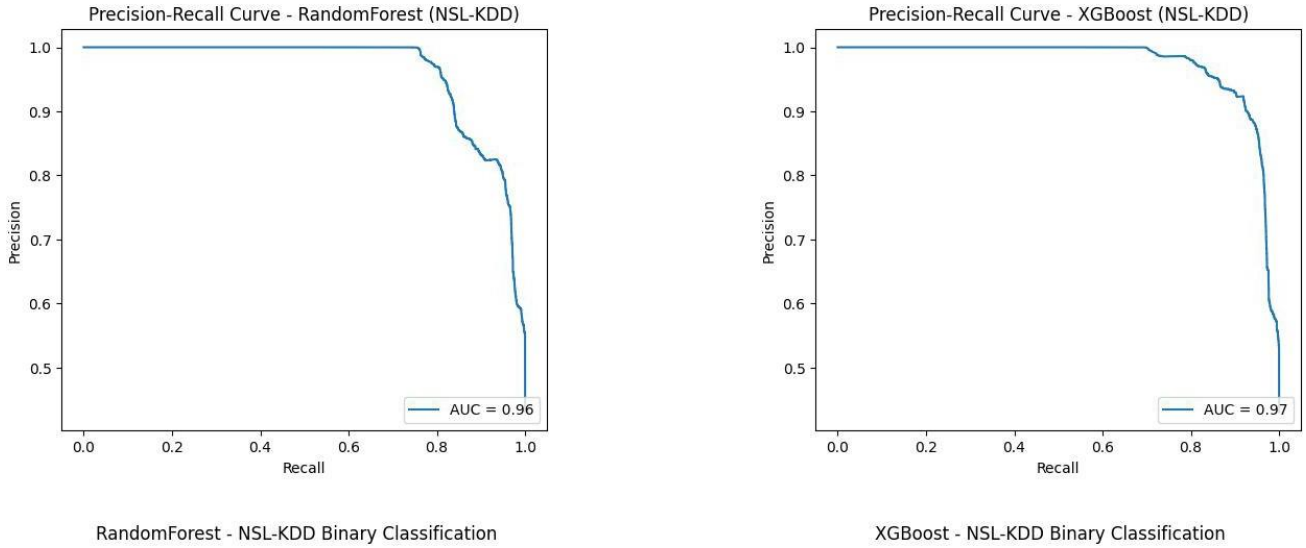


(a) RandomForest PR Curve

(b) XGBoost PR Curve

Fig. 2: Precision-Recall Curves for KDD dataset - DoS vs Normal Classification.

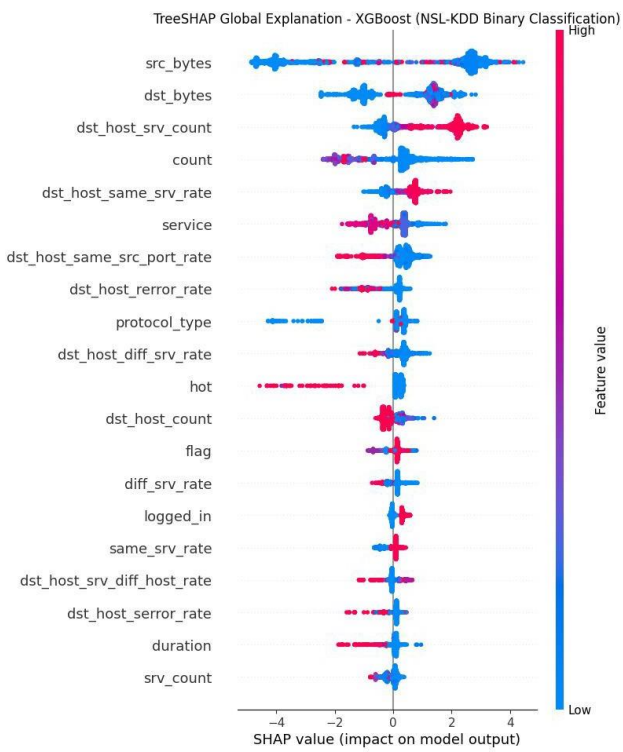


Fig. 3: XGBoost SHAP Summary: Feature importance analysis using TreeSHAP.

The confusion matrix indicates that the model correctly classified 6,074 DoS instances, while 1,093 were misclassified as Normal. Among the 15,142 Normal instances, 235 were incorrectly labeled as DoS. These results highlight the model's strong classification performance, with a high true positive rate and a well-balanced trade-off between precision and recall. The combination of the PR curve, confusion matrix, and ROC curve in Figure 4 illustrates the model's ability to effectively distinguish between normal and DoS traffic while maintaining reasonable misclassification rates.

A. LIME-Based Feature Attribution

To interpret individual classification decisions, we apply LIME, which provides a local explanation by approximating the impact of each feature on a specific instance. The classification function for an instance x in LIME can be approximated as:

$$M(x) = \sum_{i=1}^{N} w_i f_i + b, \tag{1}$$

where:

- w_i represents the weight assigned to feature f_i
- b is the model's bias term

Applying this equation to our NSL-KDD DoS vs. Normal classification, the most influential features, as seen in Fig. 5, are:

- f_1 = wrong_fragment – Fragmented packets are often associated with certain DoS attacks, such as Teardrop or Ping of Death, which exploit packet fragmentation vulnerabilities. However, in this dataset, LIME indicates that `wrong_fragment` is more frequently observed in Normal traffic (shown in blue), suggesting that the model may have learned an unexpected association between fragmentation and benign activity. This could be due to the dataset's specific distribution rather than a universal cybersecurity trend.

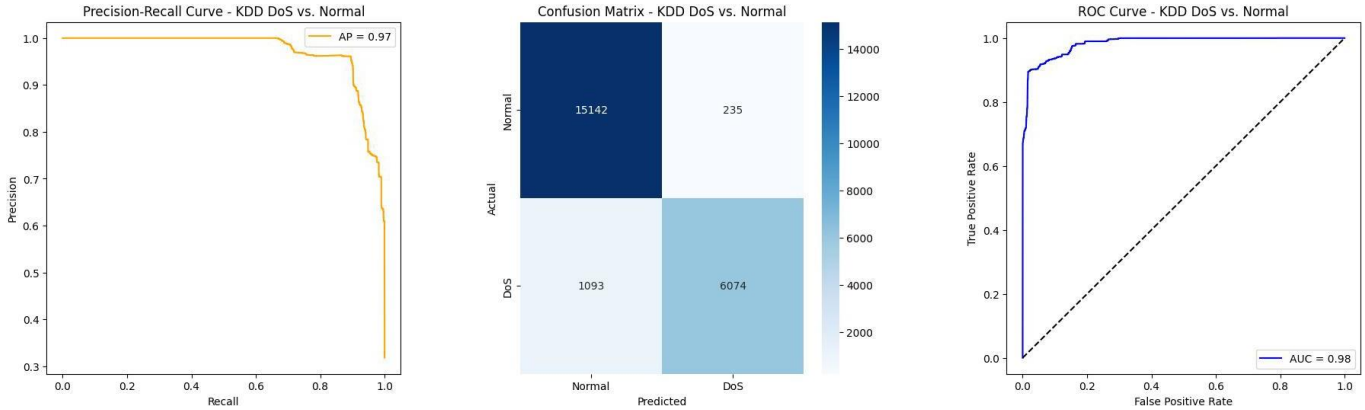


Fig. 4: Precision-Recall Curve, Confusion Matrix, and ROC Curve for KDD data - DoS vs Normal Classification.

- f_2 = src_bytes – Low source byte values suggest frequent small packet transmissions, which are indicative of SYN flood or UDP flood attacks.
- f_3 = protocol_type – The type of protocol used can differentiate attack traffic from normal traffic.
- f_4 = flag – This TCP flag helps determine the connection state and whether the packet represents a handshake, reset, or an ongoing transmission, influencing attack classification.
- f_5 = su_attempted – Failed superuser access attempts may indicate unauthorized privilege escalation, a potential sign of DoS-related behavior.
- f_6 = count – A high count value suggests frequent connection attempts to the same target, characteristic of DoS flooding behavior.
- f_7 = same_srv_rate – A low value suggests that requests are being distributed across multiple destinations rather than a single server, commonly seen in DDoS attacks.
- f_8 = urgent – The presence of urgent packets can indicate attack behavior, particularly in cases where traffic exhibits irregularities.
- f_9 = num_access_files – This feature tracks access attempts to sensitive files, which could correlate with reconnaissance activity before a DoS attack.
- f_{10} = num_shells – The number of shell access attempts, often associated with privilege escalation or unauthorized command execution.

Fig. 5 illustrates this process, showing how LIME explains the classification of a detected DoS attack by highlighting the most influential features driving the model's decision.

B. Analysis of LIME Interpretations

Fig. 5 presents a detailed breakdown of LIME feature importance. The left panel shows the model's predicted probabilities, with a high confidence level in classifying this instance as a DoS attack. The right panel visualizes the feature attributions, highlighting the impact of each network characteristic on the model's decision.

In the LIME feature importance plot, features contributing to the DoS classification are shown in orange, while those associated with Normal traffic are displayed in blue. The feature wrong_fragment had the largest negative contribution (shown in blue), indicating that its presence was strongly associated with Normal traffic in this specific instance.

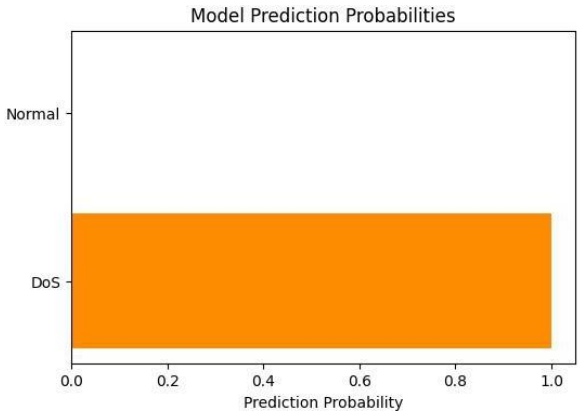
Since LIME provides only instance-based explanations, the next section will reintroduce SHAP, focusing on SHAP interaction values and LIME explanations in the UNSW-NB15 dataset. This analysis examines how feature interactions influence the model's predictions for DoS and Reconnaissance attacks, providing a comparative view between local (LIME) and global (SHAP) interpretability methods.

VII. Comparing Global (SHAP) and Local (LIME) Explanations for Normal, DoS, and Reconnaissance

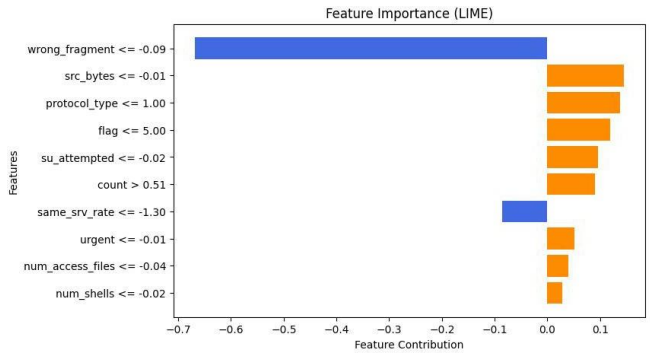
A. SHAP and LIME: Global Feature Interactions in a Multi-Class Setting

To further analyze the model's decision-making process, we examine SHAP global feature interactions within the classification scope of Normal, DoS, and Reconnaissance traffic. While previous sections explored SHAP's overall feature importance, this analysis specifically investigates how key network characteristics interact to influence predictions in a multi-class setting. LIME is also applied to highlight individual misclassifications, allowing us to compare local and global interpretability methods.

For this analysis, we trained an XGBoost classifier on the UNSW-NB15 dataset, distinguishing between Normal, DoS, and Reconnaissance traffic using a multi-class classification approach. The model performed well, as demonstrated by the Precision-Recall (PR) curves in Fig. 6, achieving AUC scores of 0.91 for DoS and 0.95 for Reconnaissance. These scores indicate that the model effectively balances precision and recall, minimizing false positives while maintaining strong classification performance.



(a) Model Prediction Probabilities (LIME)



(b) Feature Importance (LIME)

Fig. 5: LIME-Based Feature Attribution and Classification Confidence for NSL-KDD DoS vs. Normal

However, while overall performance metrics suggest high accuracy, local and global interpretability methods reveal key insights into the model's decision-making process. To efficiently compute feature interactions without processing the entire dataset, we utilized TreeSHAP on a stratified sample of 1,000 instances, ensuring a representative selection of DoS traffic while maintaining computational feasibility. The SHAP interaction analysis revealed that dur (duration), service, and proto were the top three interacting features for DoS classification within the scope of Normal, DoS, and Reconnaissance traffic.

LIME's instance-based explanation, on the other hand, highlights a misclassification case where the model incorrectly predicted Normal traffic instead of DoS, relying primarily on the proto (protocol type) feature. Additionally, other features such as sttl (time-to-live value) and ct_srv_src (count of connections to the same service from the source) contributed to the decision, suggesting that the model may have overemphasized protocol-based similarities while underutilizing stronger attack-relevant indicators. This over-reliance suggests that the model learned an incomplete decision boundary for DoS attacks, mistakenly assigning normal-like protocol values more weight than other attack-relevant indicators. SHAP's global interaction values, however, highlight dur (connection duration) and service as the most influential interacting features for distinguishing DoS attacks. This suggests that successful classification relies on understanding not only individual feature importance but also how network characteristics interact within attack patterns. By capturing these dependencies, SHAP provides a broader perspective on model decision-making across Normal, DoS, and Reconnaissance traffic. This suggests that the model should have placed greater weight on the interplay between duration, service type, and protocol rather than relying predominantly on protocol type, as observed in LIME's local explanation.

The discrepancy between LIME, which highlighted proto in a single misclassified case, and SHAP, which demonstrated the importance of feature interactions across Normal, DoS, and Reconnaissance classifications, underscores the necessity of combining local and global XAI techniques to uncover potential model biases.

To further illustrate these insights, Fig. 7 presents the global SHAP interaction values for DoS (left) and Reconnaissance (right) within the scope of Normal, DoS, and Reconnaissance classifications in the UNSW-NB15 dataset. These visualizations reveal how key features interact to influence the model's classification decisions, providing deeper insight into the relationships between attack indicators. By analyzing these interactions, we gain a broader understanding of how different attack types are distinguished, allowing for improved feature attribution and model refinement.

In addition, Fig. 8 presents the localized misclassification instance identified by LIME. The left panel of the figure displays the model's predicted probabilities, showing that the instance was misclassified with 62% confidence. The right panel illustrates the key features that contributed to this decision, revealing that the model placed significant weight on proto while failing to account for stronger interacting predictors such as dur and service, as indicated by SHAP.

This comparison between SHAP and LIME underscores the complementary nature of local and global explainability techniques in intrusion detection. While LIME effectively identifies misclassification patterns at the instance level, it may overlook broader feature interactions that influence the model's predictions. SHAP's global interaction values revealed that duration, service, and protocol collectively played a more significant role in DoS classification than protocol alone. This finding highlights a key limitation of relying solely on local explanations, as they can misrepresent the underlying decision boundaries learned by the model. By integrating both methods, we gain a more comprehensive understanding of how attack classifications are determined, allowing for better model refinement, reduced bias, and improved decision-making in real-world cybersecurity applications.

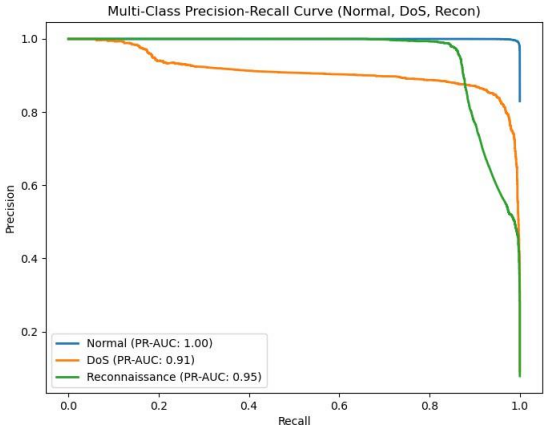


Fig. 6: Precision-Recall Curve for Multi-Class Classification (Normal, DoS, Recon) in UNSW-NB15 Dataset.

VIII. Practical Platform

We have implemented a containerized solution that allows users to interact with the trained XGBoost and Random Forest intrusion detection models beyond just training and testing, Users can query the model by selecting an index from the dataset to generate LIME-based explainability outputs or input custom feature values to probe the model's predictions. This ensures flexibility in exploring specific predictions and interpreting model decisions, making it useful for researchers and practitioners. The demonstration of this containerized solution is available on GitHub and is presented using Docker and Flask, allowing users to interact with the model in a self-contained, easily deployable environment. This approach ensures seamless access without complex setup, making it ideal for secure and scalable deployment.

IX. Conclusion

The integration of XAI techniques, such as SHAP and LIME, is essential for validating and improving the reliability of AI-driven intrusion detection systems in operational environments. By bridging the gap between model predictions and human interpretability, these techniques empower security analysts to refine detection rules, reduce false positives, and strengthen responses to evolving cyber threats. While ML models, including Random Forest and XGBoost, achieve high accuracy in classifying network attacks, their decision-making processes must be transparent and interpretable to ensure meaningful security applications in production settings.

This study demonstrates the effectiveness of incorporating both global and local explainability methods, with SHAP offering dataset-wide feature importance insights and LIME providing case-specific interpretability. Our findings underscore the significance of leveraging diverse benchmark datasets (NSL-KDD, UNSW-NB15) to comprehensively evaluate intrusion detection models. By analyzing feature interactions and misclassification patterns, we highlight the importance of balancing model interpretability with classification performance. The integration of multi-class classification and explainability techniques provides deeper insights into how different attack types are distinguished, aiding in model refinement and real-world applicability.

Through a comparative analysis of binary and multi-class classification, we highlight the importance of contextualizing attack detection at different levels of granularity. The study confirms that AI-enhanced intrusion detection, augmented by XAI tools, fosters greater trust and usability in cybersecurity applications. Our practical platform implementation demonstrates how these research findings can be effectively deployed in operational environments, providing security teams with actionable insights and improved decision-making capabilities.

Future research should focus on refining adversarial resilience to ensure robust performance against evolving threats, exploring federated learning approaches for distributed detection in large-scale networks, developing more sophisticated explainability frameworks that address operational requirements, investigating real-time performance optimization for production environments, and enhancing integration capabilities with existing security infrastructure. These advancements will further strengthen the role of AI in cyber defense while ensuring that security operations teams can effectively leverage these technologies in their daily operations.

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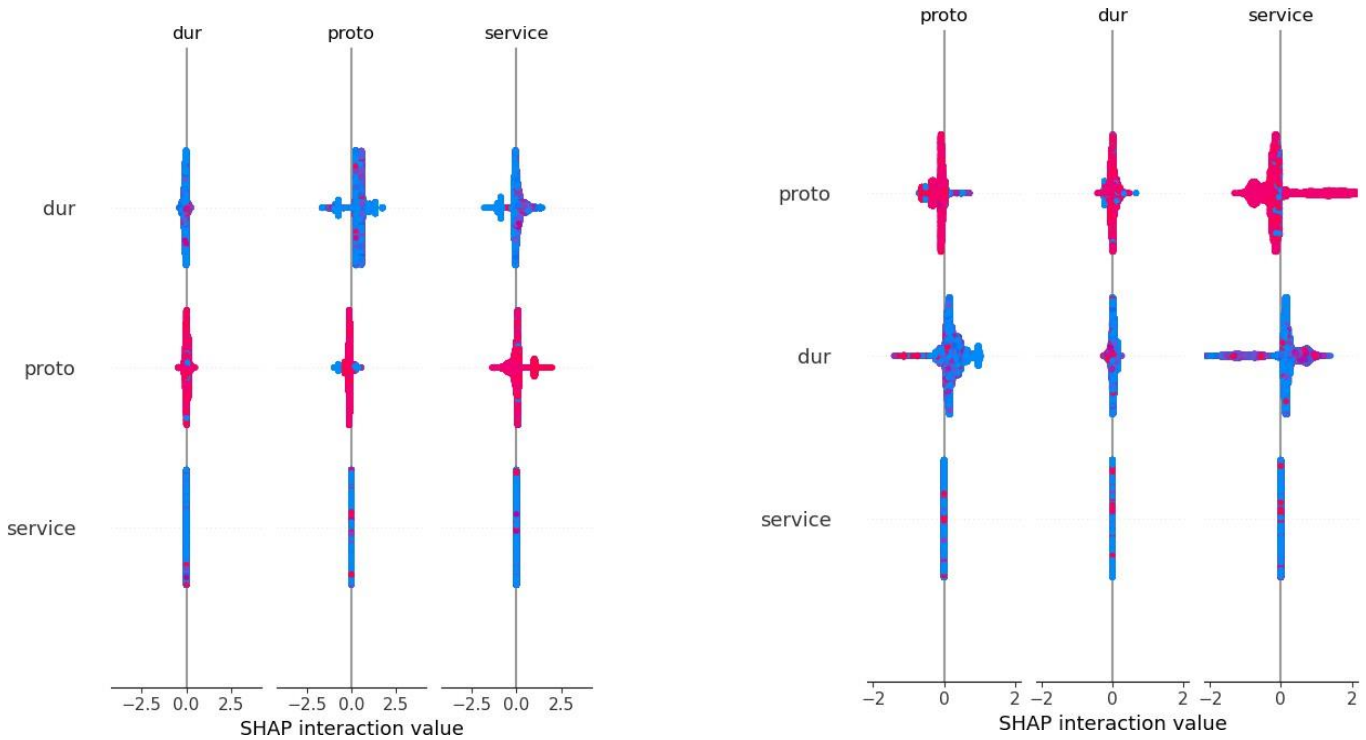

Fig. 7: SHAP Interaction Values for DoS and Reconnaissance Attacks (UNSW-NB15 Dataset).

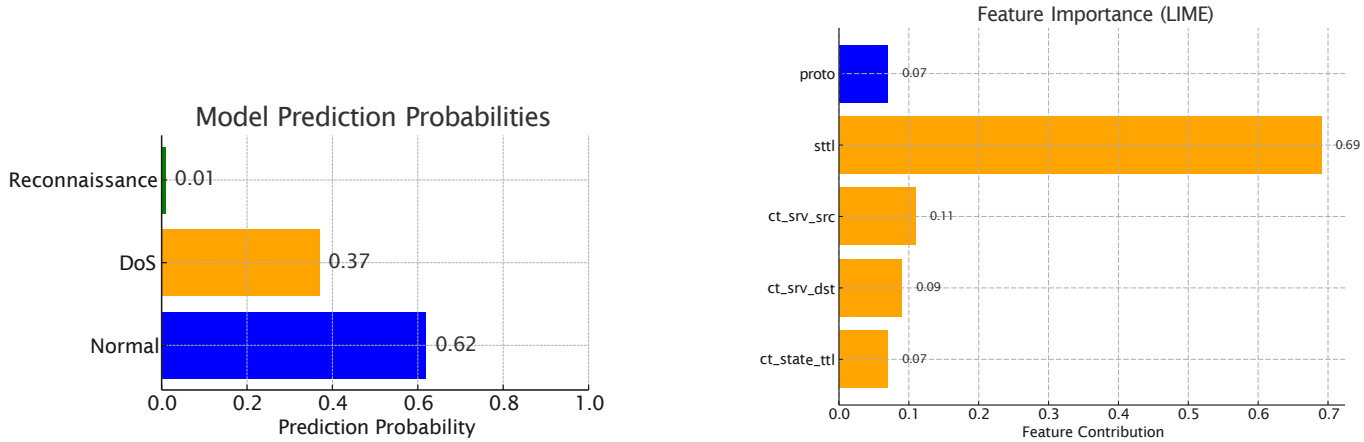

Fig. 8: LIME-Based Explainability of a Misclassified Instance in UNSW-NB15 (Normal, DoS, and Reconnaissance).

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